In JEE Advanced 2016, FIITJEE Students bag 36 in Top 100 AIR, 75 in Top 200 AIR, 183 in Top 500 AIR. 3541 Students from Long Term Classroom/ Integrated School Program & 4423 Students from All Programs have qualified in JEE Advanced, 2016

FIITJEE

JEE(Advanced)-2017

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

FULL TEST – IV
(Paper-1)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	**B**	**D**	**C**
2.	**A**	**A**	**A**
3.	**A**	**C**	**D**
4.	**A**	**C**	**D**
5.	**C**	**D**	**A**
6.	**B**	**C**	**C**
7.	**C**	**C**	**C**
8.	**B**	**B**	**D**
9.	**A**	**A**	**C**
10.	**C**	**C**	**B**
11.	**B**	**A, B, D**	**B, C, D**
12.	**B, C, D**	**A, C, D**	**A, B, C**
13.	**A, C, D**	**D**	**B, C**
14.	**C, D**	**A, B, C**	**B, D**
15.	**A, D**	**A, C, D**	**A, B, D**
1.	**2**	**4**	**4**
2.	**0**	**0**	**2**
3.	**3**	**1**	**4**
4.	**9**	**8**	**3**
5.	**2**	**2**	**2**

Physics

PART – I

SECTION – A

1. **B**

When block is further pushed by x in liquid buoyant force increases due to $\frac{4x}{3}$ height of liquid displaced as $\pi R^2 x = \left(4\pi R^2 - \pi R^2\right) y$

$\Rightarrow y = \frac{x}{3}$

Hence extra buoyant force $= \pi R^2 \left(x + \frac{x}{3}\right).2\rho g$

2. **A**

$E_{ind}.2\pi R = \pi R^2 \frac{dB}{dt} \Rightarrow E_{ind} = \frac{dB}{dt}$

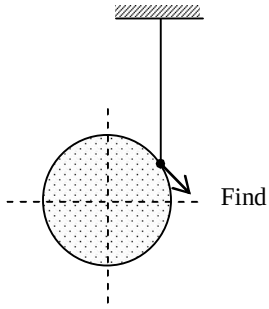
Now, $T = mg + qE\cos 60^o$

$2mg = mg + \frac{qE}{2}$

$qE_{ind}\cos 60 = mg$

$q.\frac{1}{2}.\frac{1}{2} = mg$

$\Rightarrow q = 4mg$



3. **A**

$g = -\frac{dV}{dr}$

$\frac{-K}{r} = -\frac{dV}{dr}$

$\int_{V_0}^{V} dV = K\int_{r_0}^{r} \frac{dr}{r}$

$V = K\ln\left(\frac{r}{r_0}\right) + V_0$

4. **A**

$F_1 = \frac{GMm}{4R^2}$, $m_{cavity} = \frac{M}{8}$

$F_2 = F_1 - F_{cavity} = \frac{GMm}{4R^2} - \frac{G\frac{M}{8}.m}{\left(\frac{3R}{2}\right)^2} = \frac{7}{36}\frac{GMm}{R^2}$

$\Rightarrow \frac{F_1}{F_2} = \frac{9}{7}$

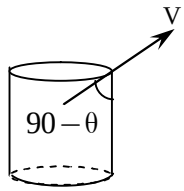
5. **C**

Effective area receiving the particles is

$$\frac{\pi R^2}{2}\sin\theta + 2Rh\cos\theta + \frac{\pi R^2}{2}\sin\theta$$

Hence, $F = \rho v^2 A_{eff.}$

$\rho = Fv = \rho v^3 A_{eff.}$



6. **B**

More the value of T more must be extension in rubber band. Tension in rubber band is in decreasing order from right to left.

7. **C**

$V_A = V_C = V_E$

Hence, only one capacitor between E and F remains effective.

So, $U = \frac{1}{2}CV^2 = \frac{1}{2}\frac{\varepsilon_0 A}{d}V^2$

8. **B**

9. **A**

Since refractive index is changing ray cannot travel in straight line. Hence (C) and (D) are wrong options. Initially it has to bend towards normal and later on away from normal.

10. **C**

For plano convex lens

$$\frac{1}{f} = \frac{1}{40} = \frac{(\mu - 1)}{R}$$

For biconvex lens

$$\frac{1}{v_1} - \frac{1}{-10} = \frac{1}{20} \Rightarrow v_1 = -20 \text{ cm}$$

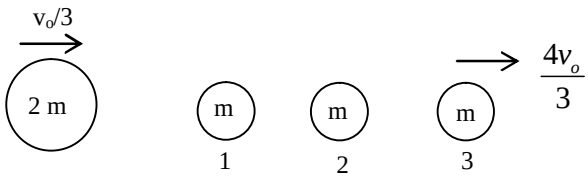
$$\frac{\mu}{\infty} - \frac{1}{-(d+20)} = \frac{(\mu-1)}{R} \Rightarrow \frac{1}{d+20} = \frac{1}{40} \Rightarrow d = 20$$

11. **B**

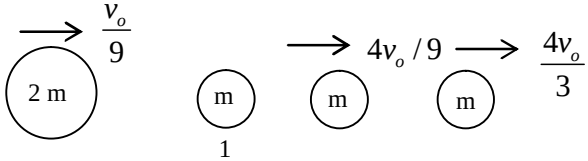
Ring will get some angular velocity about its diameter due to torque of magnetic field.

12. **B, C, D**

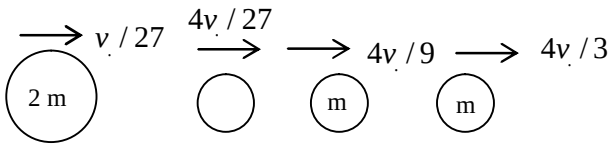
When 2m collides with ball 1 which in turn with 2 and then with 3, then



Now 2m again strikes with ball which in turn with ball 2, then



Now 2m again strikes with ball 1, then

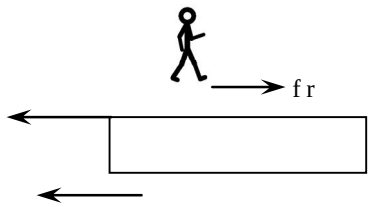


Above is the final state.

13. **A, C, D**

In frame of plank work done by friction on man is zero as w.r.t. plank there is no slipping at contact. Work done by friction on plank is positive.

Foot of person moves leftward, plank moves leftward.



14. **C, D**

Force on left and right inclined faces must have same and opposite horizontal components and force on base is $(\rho gh + P_{atm})A$.

15. **A, D**

At high temperature vibrational degrees of freedom get excited.

SECTION – C

1. **2**

In one second block comes to lower extreme and after switching on the electric field equilibrium of block is with natural length of spring.

2. **0**

At highest point, $q\bar{v} \times \bar{B} = mg \Rightarrow$ particle does not come back, hence $T = \infty$

$qvB = mg$

$q.100.B = mg$

3. **3**

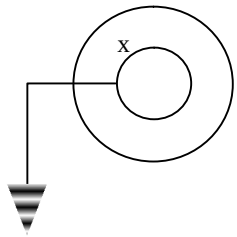
Let charge on inner shell is x.

Then,

$$V_{\text{inner shell}} = \frac{Kq}{3r} + \frac{Kx}{r} = 0$$

$$\Rightarrow x = \frac{-q}{3}$$

So, charge flowing through switch is $\frac{q}{3}$.



4. **9**

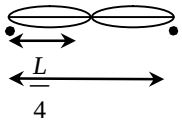
$\Delta Q = 15000\ J = nC_P\Delta T\ \&\ \Delta U = nC_V\Delta T$

$$\frac{\Delta U}{\Delta Q} = \frac{C_V}{C_P} = \frac{1}{\gamma} \Rightarrow \Delta U = \frac{\Delta Q}{\gamma} \Rightarrow \Delta U = \frac{3}{5} \times 15 = 9\ KJ$$

5. **2**

$$\frac{5\lambda}{2} = \ell \Rightarrow \lambda = 6$$

$$f = \frac{v}{\lambda} = \frac{12}{6} = 2\ Hz$$



Chemistry

PART – II

SECTION – A

1. **D**

Lime removes temporary hardness & boiling also removes temporary hardness.

$CaCl_2 + Na_2CO_3 \longrightarrow CaCO_3\downarrow + Na_2CO_3$ & $Ca(HCO_3)_2 + Na_2CO_3 \longrightarrow CaCO_3\downarrow + NaHCO_3$

2. **A**

Use law of gram equivalence.

3. **C**

$AgCNS \overset{H_2O}{\rightleftharpoons} \underset{S+S'}{Ag^+} + \underset{S}{\bar{S}}CN \Rightarrow K_{sp} = 8.5 \times 10^{-12} = (S+S')S$ ..... (1)

$AgBr \rightleftharpoons \underset{S+S'}{Ag^+} + \underset{S}{Br^-} \Rightarrow 5 \times 10^{-13} = (S+S')S$ ..... (2)

Adding (i) and (ii),

$(S+S')^2 = 9\times10^{-12}$

$(S+S') = 3\times10^{-6}$

4. **C**

Methyl is electron releasing group. Also refer ortho effect.

5. **D**

$AgNO_3 \xrightarrow{Na_2S_2O_3} Ag_2S_2O_3 \xrightarrow{H_2O} \underset{\text{Black}}{Ag_2S\downarrow}$

6. **C**

$C_6H_{12}O_6 \xrightarrow[\Delta]{conc H_2SO_4} C + H_2O$

7. **C**

Refer rules of IUPAC.

8. **B**

PR_3 is π acid ligand which forms π bond with C.M using d orbital of P.

9. **A**

$\text{Efficiency} = \dfrac{\Delta G^0}{\Delta H^0}\times100 = 85$

$\Rightarrow \Delta G^0 = -\dfrac{85\times285\times10^3}{100}$

$= -85\times2850\,kJ$

$= -2\times96500\times E^0_{cell}$

$E^0_{cell} = 1.26$ Volt

10. **C**

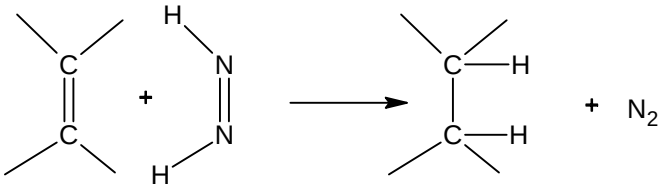
A is C_2H_5CHO and B is Me_2S.

Me_2S can be used in reductive ozonolysis.

$CH_2=CH_2 \xrightarrow[ii)\ Me_2S]{i)\ O_3} 2HCHO + Me_2SO$

11. **A, B, D**

$N_2H_4 + H_2O_2 \rightarrow N_2H_2 + 2H_2O$



12. **A, C, D**

Hexagonal, rhombohedral and triclinic have only one bravais lattice.

13. **D**

Anhyd. $CuSO_4$ is white, while $CuSO_4.5H_2O$ is blue, other complexes (A) and (B) are also coloured.

14. **A, B, C**

A & B are peroxide & hydroperoxide which oxidise Fe^{2+} to Fe^{3+} and these peroxides and hydroperoxides are explosives.

15. **A, C, D**

Proline has 2° amino group, which can not be diazotised.

SECTION – C

1. **4**

$$H_3C-\underset{\underset{OH}{|}}{CH}-CH_3 + 4I_2 + 6\bar{O}H \rightarrow CHI_3 + CH_3COO^- + 5I^- + 5H_2O$$

2. **0**

Among cubic unit cells only FCC has O.V.

3. **1**

For 10 ml standard gold solution starch required is 0.02 gm, i.e., 20 mg

So, gold number is $20 = 2 \times 10^1$

4. **8**

$$\frac{\Delta H_2 - \Delta H_1}{T_2 - T_1} = \Delta C_p$$

$$\Rightarrow \Delta H_2 = -28.58 \times 10^3 + 32 \times 60$$

$$= -28580 + 1920 = -26660 \text{ J}$$

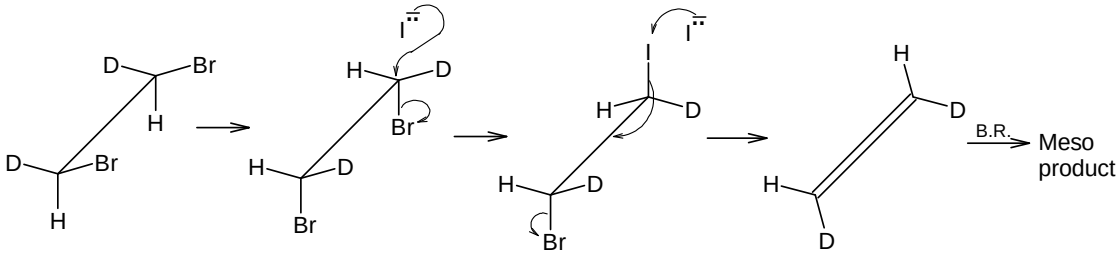
i.e., $\Delta H_{87^oC} = -26.66$ KJ

Efficiency of fuel cell $= \frac{\Delta G}{\Delta H} \times 100 = 90 \Rightarrow \Delta G = \frac{\Delta H \times 90}{100}$

$\Delta G = \frac{-26.66 \times 90}{100} = -23.994 \approx -24$ KJ/mol $\Rightarrow 3 \times a \Rightarrow a = 8$

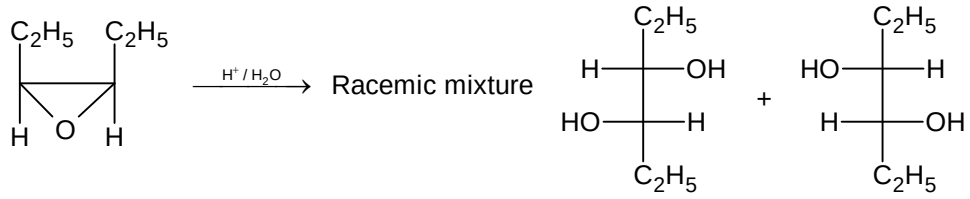
5. **2**

(a) Different mechanism operates with 1° vicinaldihalides, and first cis alkene then meso product are obtained here.

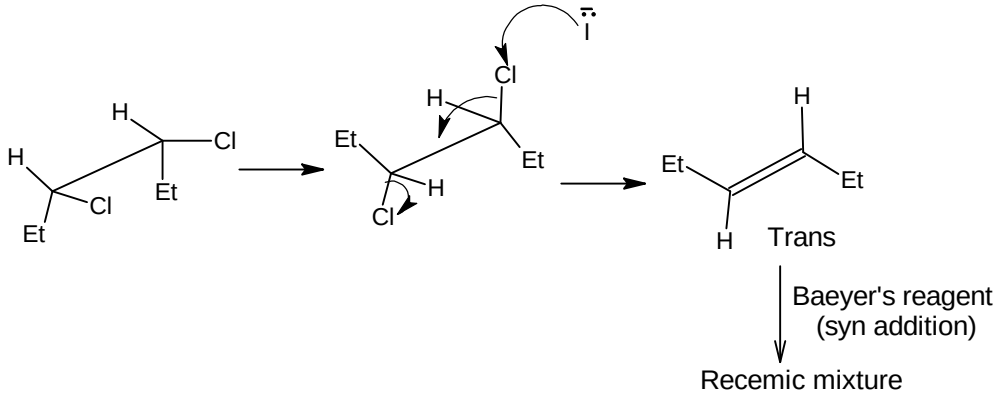


(b) Trans alkene $\Rightarrow$ syn addition $\Rightarrow$ racemic mixture product, as epoxide is formed.

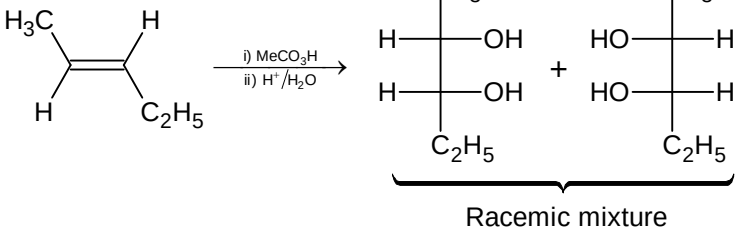
(c)



(d)



(e)



(f) Racemic mixture is obtained. Meso can not be obtained in this reaction as alkene is not symmetrical.

(g) First cis alkene, then sys addition of –OH group, so meso product.

(h) Anti addition, but alkene is not symmetrical, product is racemic mixture.

Mathematics

PART – III

SECTION – A

1. **C**

$\because I = \int_0^{\pi} f(\tan 2x)\cos x\, dx$ ..... (1)

$I = \int_0^{\pi} f(-\tan 2x)(-\cos x)dx = I$

$\Rightarrow I = 2\int_0^{\pi/2} f(\tan 2x)\cos x\, dx$

$\Rightarrow I = 2\int_0^{\pi/2} f(-\tan 2x)\sin x\, dx$

$2I = 2\int_0^{\pi/2} f(\tan 2x)(\cos x - \sin x)dx$

$= \sqrt{2}I\int_0^{\pi/2} f(\tan 2x)\cos\left(\frac{\pi}{4} + x\right)dx$

$= \sqrt{2}\int_{-\pi/4}^{\pi/4} f(-\cot 2x)(-\sin x)dx$

$I = 2\sqrt{2}\int_0^{\pi/4} f(\cot 2x)\sin x\, dx$

$= 2\sqrt{2}J$

2. **A**

$a_{11} = f\left(x^{2-1-2}\right) = f\left(\frac{1}{x}\right)$, $a_{21} = f\left(x^{4-1-2}\right) = f(x)$, $a_{12} = f\left(x^{2-2-2}\right) = f\left(\frac{1}{x^2}\right)$ and $a_{22} = f\left(x^{4-2-2}\right) = f(1)$.

$\therefore A = \begin{pmatrix} f\left(\frac{1}{x}\right) & f\left(\frac{1}{x^2}\right) \\ f(x) & f(1) \end{pmatrix}$ and $B = \begin{pmatrix} f(1) & f\left(\frac{1}{x}\right) \\ f\left(x^2\right) & f(x) \end{pmatrix}$

$\therefore A + B^T = \begin{pmatrix} f\left(\frac{1}{x}\right) + f(1) & f\left(\frac{1}{x^2}\right) + f\left(x^2\right) \\ f(x) + f\left(\frac{1}{x}\right) & f(1) + f(x) \end{pmatrix}$

From given matrix,

$f\left(\frac{1}{x}\right) + f(1) = -x^2 f\left((x) + f(1)\right)$ ..... (i)

$f\left(\frac{1}{x^2}\right) + f\left(x^2\right) = \left(1 - \frac{1}{x^4}\right) f\left(\frac{1}{x^2}\right)$ ..... (ii)

$f(x)+f\left(\frac{1}{x}\right)=f\left(\frac{1}{x}\right)\left(1-\frac{1}{x^2}\right)$ ..... (iii)

$f(1)+f(x)=-\frac{1}{x^2}f\left(\frac{1+f(1)}{x}\right)$ ..... (iv)

From (ii), $x=1 \qquad 2f(1)=0 \Rightarrow f(1)=0$

$\therefore$ Eqn. (i), $-x^2 f(x)=f\left(\frac{1}{x}\right)$ ..... (*)

Now, $\int_x^{1/x} f(t)\,dt = I$

Replace $t \to \frac{1}{t}$

$$\Rightarrow I=\int_{1/x}^{x} f\left(\frac{1}{t}\right)\left(-\frac{1}{t^2}\right)dt=\int_x^{1/x} f\left(\frac{1}{t}\right)\frac{1}{t^2}dt$$

$$\Rightarrow 2I=\int_x^{1/x}\left(f(t)+\frac{1}{t^2}f(t)\right)dt=0 \text{ from (*)}$$

i.e., $I=0=f(1)$

Option (a) is correct.

3. **D**

$$\left[f(x)\quad g(x)\quad h(x)\right]=\begin{vmatrix}\cos x & \sin x & \cos 2x\\ \tan x & \sin 3x & \cos 4x\\ \cos 3x & \sin 5x & \cos 6x\end{vmatrix}$$

$$\therefore \int_0^{\pi}[f,\ g,\ h](x)\,dx=\int_0^{\pi}[f,\ g,\ h](\pi-x)\,dx$$

$$\Rightarrow \int_0^{\pi}[f,\ g,\ h]\,dx=-\int_0^{\pi}[f,\ g,\ h]\,dx \qquad (\because C_1 \to -C_1)$$

$$\Rightarrow \int_0^{\pi}[f,\ g,\ h]\,dx=0$$

Option (d) is correct.

4. **D**

$\vec{a}_1=(a,\ b),\ \vec{a}_2=\left(a-2,\ b-\sqrt{5}\right),\ \vec{a}_3=\left(a+2,\ b-\sqrt{5}\right)$

Area of Δ, whose vertices are $\vec{a}_1,\ \vec{a}_2$ and $\vec{a}_3$, is

$$\Delta=\frac{1}{2}\begin{vmatrix}1 & a & b\\ 1 & a-2 & b-\sqrt{5}\\ 1 & a+2 & b-\sqrt{5}\end{vmatrix}$$

$$\Rightarrow \Delta^2 = \frac{1}{4}\begin{vmatrix} 1 & a & b \\ 1 & a-2 & b-\sqrt{5} \\ 1 & a+2 & b-\sqrt{5} \end{vmatrix}\begin{vmatrix} 1 & a & b \\ 1 & a-2 & b-\sqrt{5} \\ 1 & a+2 & b-\sqrt{5} \end{vmatrix} \text{ and}$$

$$\left|\vec{a}_1 - \vec{a}_2\right| = 3,\ \left|\vec{a}_1 - \vec{a}_3\right| = 3,\ \left|\vec{a}_1 - \vec{a}_3\right| = 4$$

$$\Rightarrow \Delta = \sqrt{5\times 2\times 2\times 1} = 2\sqrt{5}$$

$$\left(2\sqrt{5}\right)^2 = \frac{1}{4}\begin{vmatrix} 1+\vec{a}_1.\vec{a}_1 & 1+\vec{a}_1.\vec{a}_2 & 1+\vec{a}_1.\vec{a}_3 \\ 1+\vec{a}_2.\vec{a}_1 & 1+\vec{a}_2.\vec{a}_2 & 1+\vec{a}_2.\vec{a}_3 \\ 1+\vec{a}_3.\vec{a}_1 & 1+\vec{a}_3.\vec{a}_2 & 1+\vec{a}_3.\vec{a}_3 \end{vmatrix}$$

$\therefore$ Required answer = 4 x 5 x 4 = 80

Option (d) is correct.

5. **A**

$$I = \int_0^{\pi/2} \frac{\sin^3 x\cos x}{\cos x+\sin x}dx$$

$$I = \int_0^{\pi/2} \frac{\cos^3 x\sin x dx}{\sin x+\cos x} \qquad \text{Prop. } \int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$$

$$\Rightarrow 2I = \int_0^{\pi/2} \frac{\sin x\cos x\ dx}{\sin x+\cos x}$$

$$2I = 2\int_0^{\pi/4} \frac{\sin x\cos x dx}{\sqrt{2}\cos\left(x-\frac{\pi}{4}\right)}$$

$$I = \frac{1}{2\sqrt{2}}\int_0^{\pi/4} \frac{\sin 2x}{\cos\left(x-\frac{\pi}{4}\right)}dx = \frac{1}{2\sqrt{2}}\int_0^{\pi/4} \frac{\cos 2x}{\cos x}dx$$

$$= \frac{1}{2\sqrt{2}}\int_0^{\pi/4} (2\cos x - \sec x)\,dx = \frac{1}{2\sqrt{2}}\left[2\sin x - \ln\left|\sec x+\tan x\right|\right]_0^{\pi/4}$$

$$= \frac{1}{2\sqrt{2}}\left[2\left(\frac{1}{\sqrt{2}}\right) - \ln\left|\sqrt{2}+1\right|\right] = \frac{1}{2} - \frac{1}{2\sqrt{2}}\ln\left|\sqrt{2}+1\right|$$

Option (a) is correct.

6. **C**

$\because \left|f(n)\right| \le |0| \Rightarrow f(n) = 0$

& $\left|f(x)-f(n)\right| \le |\sin \pi x|$

$\Rightarrow \left|f'(n)\right| \le \left|\lim_{x\to n}\cos \pi x \times \pi\right|$

$\Rightarrow \left|f'(n)\right| \le \pi$

7. **C**

$\because f(x) = \tan 2x - \tan x$

$f(1) = \tan 2 - \tan 1$

$f(2) = \tan 4 - \tan 2$

.....

.....

$f(2^n) = \tan 2.2^n - \tan 2^n$

$g(x) = \tan 2^{x+1} - \tan 1$

8. **D**

Using $\tan\theta - \tan\frac{\theta}{2} = \tan\frac{\theta}{2} + \tan\theta\tan^2\frac{\theta}{2}$

9. **C**

$\because \quad \beta = \frac{f(4)+f(12)}{2}$

$= \frac{4+12}{2} = 8$

10. **B**

$f^{-1}(x) = x - 1$ then $h^{-1}(x) = (g \cdot f(x))^{-1}$

$\therefore \frac{d}{dx}(h^{-1}(x)) = 3x^2 + 1 \qquad = f^{-1} \cdot g^{-1}(x)$ (reversal law)

$= f^{-1}(x^3 + x^3 + x + 1)$

$= x^3 + x$

$(h^{-1}(1))' = 3 + 1 = 4$

11. **B, C, D**

$$\int \frac{2x^2 - \frac{1}{x} + \sqrt{x}}{(1+x^3)}dx = \int \frac{x\left(2x - \frac{1}{x^2}\right)dx}{x\left(x^2 + \frac{1}{x}\right)} + \int \frac{\sqrt{x}dx}{1+\left(x^{3/2}\right)^2}$$

$$= \ln\left(x^2 + \frac{1}{x}\right) + \frac{2}{3}\tan^{-1}\left(x^{3/2}\right)$$

$$\Rightarrow \int_1^2 \frac{2x^2 - \frac{1}{x} + \sqrt{x}}{(1+x^3)}dx = \ln\left(\frac{9}{2}\right) - \ln 2 + \frac{2}{3}\cos^{-1}\frac{1}{3} - \frac{\pi}{6} = \ln\left(\frac{9}{4}\right)$$

Options (b), (c), (d) are correct.

12. **A, B, C**

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

$\Rightarrow a_{11} + 2a_{12} = a_{11} + a_{21}$ ..... (i)

$a_{11} + a_{12} = a_{12} + a_{22}$ ..... (ii)

$a_{21}+2a_{22}=2a_{11}+a_{21}$ ……. (iii)

and $\Rightarrow a_{21}+a_{22}=2a_{12}+a_{22}$ ……. (iv)

From (ii), $a_{11}=a_{22}$

From (iv), $a_{21}=2a_{12}$

$\because a_{11}a_{22}=a_{21}a_{21}\Rightarrow (a_{11})^2=2(a_{12})^2 \Rightarrow a_{11}=\sqrt{2}a_{12}$

13. **B, C**

Using $f(x)=\frac{\ln x}{x}$ has two distinct roots.

14. **B, D**

$$\Delta(x)=\begin{vmatrix} f & g & h \\ f' & g' & h' \\ f'' & g'' & h'' \end{vmatrix}$$

$$\Rightarrow \Delta'(x)=\begin{vmatrix} f & g & h \\ f' & g' & h' \\ f''' & g''' & h''' \end{vmatrix}$$

15. **A, B, D**

Confocalconics always intersect orthogonally & circle touches at vertex of ellipse .

$\therefore k=1$

SECTION – C

1. **4**

$$I=\int_1^2 \frac{x^3-3x^2}{\ln(x-1)} \text{ and } J=\int_2^3 \frac{3x-5}{\ln(x-2)}$$

$$\Rightarrow J=\int_1^2 \frac{3x-2}{\ln(x-1)}$$

$$J+I=\int_1^2 \frac{x^3-3x^2+3x-2}{\ln(x-1)}=\int_1^2 \frac{(x-1)^3-1}{\ln(x-1)}dx$$

$$=\int_1^2 \frac{(x-1)^3-1}{\ln(x-1)}dx=\int_0^1 \frac{x^3-1}{\ln x}dx=\ln 4$$

$\therefore e^{\ln 4}=4$

2. **2**

$\overrightarrow{OA}=a\hat{i}+b\hat{j}+c\hat{k}$ and $a^2+b^2+c^2=4$

$$\Rightarrow AB=\left(25-24\cos^2\frac{\theta}{2}\right)$$

$$\therefore |QD|=\frac{2}{5}\cos\frac{\theta}{2}$$

$$\Rightarrow 25|QD|=2$$

3. **4**

$$\frac{3x^2+1}{\sqrt{x^4+x^2}}=\frac{3x^2+1}{x\sqrt{x^2+1}}=\frac{x^2+1+2x^2}{x\sqrt{x^2+1}}$$

$$=\frac{\sqrt{x^2+1}}{x}+\frac{2x}{\sqrt{x^2+1}}\geq 2\sqrt{2}$$

$$f(x)=2\sqrt{2}$$

$$\therefore x^2+1=2x^2$$

$$x^2=1$$

$$x=\pm 1 \qquad x=1$$

$$\int_0^{\sqrt{2}}\frac{\left(3x^2+1\right)dx}{\sqrt{x^4+x^2}}\geq 2\sqrt{2}\times\sqrt{2}=4$$

4. **3**

$$\because \left|\text{adj}(A)\right|=|A|^{n-1}$$

$$\Rightarrow \left|\text{adj}\left(\text{adj}(A)\right)\right|=\left|\text{adj}(A)\right|^{n-1}=|A|^{(n-1)^2}$$

$$\Rightarrow \left|\text{adj}\left(\text{adj}\left(\text{adj}(A)\right)\right)\right|=|A|^{(n-1)^3}\Rightarrow (n-1)^2=4$$

$$n-1=2$$

$$n=3$$

5. **2**

Given $a_{11}=b_{11}$, $a_{22}=b_{22}$, $a_{12}=-b_{12}$, $a_{21}=-b_{21}$

and $AB=BA=I$

$$\Rightarrow A=\begin{pmatrix}a_{11} & a_{12}\\ a_{21} & a_{11}\end{pmatrix} \text{ and } |A|=1$$

$$\therefore a_{11}^2-a_{12}a_{21}=1$$

$$a_{12}a_{21}=a_{11}^2-1$$

$$\Rightarrow a_{12} \text{ or } a_{21}=(a_{11}-1) \text{ or } (a_{11}+1)$$

$$\Rightarrow |a_{12}-a_{21}|=2$$